

Aniruddha Chowdhury, Ph.D., E.I.T

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[LinkedIn](#) | [GitHub](#) | Do not require sponsorship | Willing to relocate

Summary

Mechanical Engineer E.I.T. (P.E. passed in HVAC and Refrigeration) with 5 years of industrial experience in transit bus manufacturing and chemical processing. Certified Lean Manufacturing Specialist and Six Sigma Green Belt with deep knowledge in AutoCAD MEP, HVAC system design, load calculation, process optimization.

Education

Ph.D. in Engineering | Louisiana Tech University | CGPA: 3.85/4.00

Master's in Mechanical Engineering | Louisiana Tech University | CGPA: 3.92/4.00

Bachelor's in Mechanical Engineering | National Institute of Technology, Jamshedpur, India | CGPA: 3.49/4.00

Technical Skills

Computational: MATLAB, Python, SQL, Power BI, Tableau, Minitab, BASH.

Molecular Dynamics: LAMMPS, Visual Molecular Dynamics (VMD), Ovito, MarvinStudio.

FEA Software: SolidWorks, AutoCAD MEP, Revit MEP.

Computational tools and programming: MATLAB, Revit, HAP V6, MS Excel, LAMMPS, SolidWorks, AutoCAD, CATIA, SQL, Power BI, Tableau, Minitab, Aspen Plus.

Code and Standards: ASHRAE 90.1, ASHRAE 62.1, ASHRAE 55; NFPA 820, NFPA 90A, Quality (ISO 9001, ISO 13043, ISO 14001), Automotive (SAE J639, SAE J2670).

Professional Experience

Researcher, Louisiana Tech University, Ruston, LA | March 2021 – May 2026

- Directed collaborative research activities across cross-functional teams for a multi-year federally funded project.
- Led independent research initiatives and mentored graduate students on thermochemical and data analysis.
- Effectively communicated research findings to broader scientific community through 3 publications, 8 conference and symposium proceedings.
- Engaged with asphalt industry partners for adoption of modeling frameworks as high-level screening tool by professionals.

Process Engineer, Al Razi Chemical Complex Limited, Bangladesh | Jan. 2018 – Dec. 2020

- Modeled a hydrogen peroxide production process in Aspen Plus to perform material and energy balance analysis.
- Received a Kaizen Award for improving process capability of a distillation unit (Cpk from ~1.14 to ~1.4).
- Six Sigma Green Belt with extensive experience in Distributed control system monitoring and optimization.
- Supervised 25+ operators and oversaw standard operating procedures, and safe chemical handling practices.
- Prepared daily production and shift reports, tracked KPIs, and shared operational issues and recommendations with senior engineers.

Assistant Mechanical Engineer, AFTAB Automobiles Limited, Bangladesh | Aug 2016 – Dec 2017

- Validated bus HVAC sub-assemblies via leak, pressure-integrity, and airflow testing, releasing ~30 units/month at 100% first-pass yield.
- Supervised quality validation of 3,000+ injection-molded bus seat assemblies to automotive safety standards.
- Directed FRP compression-molding quality for exterior cowlings and interior structures, owning inspections at start-up, changeovers, and shift handoffs.
- Modeled vehicle chassis and prototype components in SolidWorks and AutoCAD, from client spec to released drawings.
- Translated client specifications into technical Bill of Materials (BOM) for raw materials, authored tender bidding documents, and prepared detailed financial expenses reports for clients.

Professional Certificates & License

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| • P.E. passed (Mechanical- HVAC & Refrigeration) | • Six Sigma Green Belt |
| • Engineer in Training (Louisiana) EI.0036277 | • Lean Manufacturing Certification |
| • CSWA | • Project Management Professional (PMP) Training |

Awards

Best Presenter, Oral presentation, Graduate Research Symposium, Louisiana Tech University, 2023, 2024, 2025.

Outstanding Graduate Student Award, College of Engineering and Science, Louisiana Tech University, 2025.

Harry J Gaston, Jr Memorial (COES) Scholarship, 2025, Adamantios K. Gorgolis COES Scholarship, 2025.